

Lab 05 — Hands-on lab: from JDK to JRE, from Node to nginx

Goal: build the same API twice — single-stage, then multi-stage — and measure the difference; do the same for an "Angular" front end; then try out build caches and an image with no shell.

Prerequisites — Lab 04 completed: `~/labo-docker/04/Api.java` exists, and the images `eclipse-temurin:21-jdk` and `eclipse-temurin:21-jre-alpine` are present.

Files provided (`files/`) - `web/package.json` and `web/src/index.html` — a fake Angular project. A `cp` will stand in for the "build": we mimic the **shape** of a front-end project, not its content.

You will write every Dockerfile yourself — that is the exercise.

Step 1 — The "all-in-one" image

```
mkdir -p ~/labo-docker/05 && cd ~/labo-docker/05
cp ~/labo-docker/04/Api.java .
```

Create `Dockerfile.mono` — compilation **and** execution in the same image:

```
FROM docker.io/library/eclipse-temurin:21-jdk
WORKDIR /app
COPY Api.java .
RUN mkdir -p build && javac -d build Api.java && jar --create --file api.jar --main-class Api -C build .
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "/app/api.jar"]
```

```
podman build -f Dockerfile.mono -t api-mono:1.0 .
podman images --format 'table {{.Repository}}\t{{.Tag}}\t{{.Size}}' | grep -E 'api-
mono|temurin'
```

Observe `localhost/api-mono 1.0 488 MB` — exactly the size of `eclipse-temurin:21-jdk`. The 2 KB JAR added nothing; the JDK accounts for all of it.

```
podman run --rm --entrypoint sh api-mono:1.0 -c 'ls /app; javac -version'
```

Observe `Api.java`, `api.jar`, `build`, and `javac 21.0.x`: both the source code **and** the compiler sit in the production image.

Explanation. This image works perfectly, and that is exactly what makes it dangerous: nothing warns you that 480 MB of tooling and your sources ship with every deployment.

Step 2 — Multi-stage

Create `Dockerfile`:

```
# ----- stage 1: build -----
FROM docker.io/library/eclipse-temurin:21-jdk AS build
WORKDIR /src
COPY Api.java .
RUN mkdir -p build && javac -d build Api.java && jar --create --file api.jar --main-class Api -C build .

# ----- stage 2: runtime -----
FROM docker.io/library/eclipse-temurin:21-jre-alpine
WORKDIR /app
COPY --from=build /src/api.jar /app/api.jar
USER 1000:1000
EXPOSE 8080
ENTRYPOINT ["java","-jar","/app/api.jar"]
```

```
podman build -t api-multi:1.0 .
```

Observe the prefixes `[1/2] STEP 1/4 ...` then `[2/2] STEP 1/6 ...`: Buildah numbers the stages, and only the last one ends with `COMMIT api-multi:1.0`.

```
podman images --format 'table {{.Repository}}\t{{.Tag}}\t{{.Size}}' | grep -E 'api-mono|api-multi'
podman run --rm --entrypoint sh api-multi:1.0 -c 'ls /app; javac -version'
podman run --rm --entrypoint ls api-multi:1.0 /src
```

Observe `209 MB` versus `488 MB`, then `api.jar` on its own, `sh: javac: not found`, and `ls: cannot access '/src': No such file or directory`.

Explanation. The `build` stage lived just long enough to compile, then was thrown away. The only thing the final image knows about it is the file that `COPY --from` brought over. No sources, no JDK, no `/src` folder: none of them were ever part of its layers.

```
podman history api-multi:1.0 --format 'table {{.Size}}\t{{.CreatedBy}}' | head -6
```

Observe a `4.61kB` layer for the `COPY` of the JAR: everything else comes from the base image.

PITFALL

`COPY --from=build /src /app` would have copied the whole folder, `Api.java` and `build/` included. Copy **the artefact**, not the working directory. This is question 3.

Step 3 — Looking inside a stage

A discarded stage cannot be inspected... unless you ask the build to stop there:

```
podman build --target build -t api-build-stage .
podman run --rm api-build-stage ls -la /src
```

Observe `Api.java`, `api.jar`, `build/`: the exact content of the stage at the moment stage 2 copied `api.jar` out of it.

Explanation. `--target` is the diagnostic tool for multi-stage builds. When a `COPY --from` fails with `no such file or directory`, build the stage on its own and look inside instead of guessing paths (question 12).

```
podman rmi api-build-stage
```

Step 4 — The front end: Node builds, nginx serves

```
mkdir -p web && cp -r <lab-path>/files/web/* web/  
ls -R web
```

Create `web/Dockerfile`:

```
FROM docker.io/library/node:22-alpine AS build  
WORKDIR /app  
COPY package.json .  
RUN echo "npm ci (simulated)"  
COPY src ./src  
RUN mkdir -p dist/web/browser && cp src/index.html dist/web/browser/ && echo "ng build  
(simulated)"  
  
FROM docker.io/library/nginx:alpine  
COPY --from=build /app/dist/web/browser /usr/share/nginx/html  
EXPOSE 80
```

```
podman build -t web-multi:1.0 web  
podman images --format 'table {{.Repository}}\t{{.Tag}}\t{{.Size}}' | grep -E 'web-  
multi|node|nginx'
```

Observe `web-multi 1.0 64.2 MB` — exactly the size of `nginx:alpine` — while `node:22-alpine` weighs `167 MB`.

```
podman run -d --name web -p 18081:80 web-multi:1.0  
curl -s localhost:18081/  
podman rm -f -t 0 web
```

Observe your `index.html`, served by nginx. Also open `http://localhost:18081/` in the Windows browser.

Explanation. Node did the "building" (here, a `cp` stands in for `ng build`), then vanished. In production, the front end is a static file server — which is why the `web` image of an Angular stack never contains Node.

→ ANGULAR

On a real project, replace `RUN echo "npm ci (simulated)"` with `RUN npm ci`, replace the `cp` with `RUN npm run build`, and copy `dist/<project-name>/browser`. The `COPY package*.json → npm ci → COPY . .` split is the cache split from lab 04: the 900 MB of `node_modules` are re-downloaded only when `package-lock.json` changes.

Step 5 — A cache that survives builds

Create `Dockerfile.cache`:

```
FROM docker.io/library/eclipse-temurin:21-jdk AS build
WORKDIR /src
COPY Api.java .
RUN --mount=type=cache,target=/root/.m2 sh -c 'echo "dep-$(date +%s)" >> /root/.m2/marker; cat
/root/.m2/marker' \
&& mkdir -p build && javac -d build Api.java

FROM docker.io/library/alpine
COPY --from=build /src/build /app
```

The `marker` file simulates the Maven `~/.m2` repository: every build appends one line to it.

```
podman build --no-cache -f Dockerfile.cache -t cache-demo . 2>&1 | grep dep-
podman build --no-cache -f Dockerfile.cache -t cache-demo . 2>&1 | grep dep-
```

Observe one `dep-...` line on the first build and **two** on the second — even though `--no-cache` rebuilt everything. The `/root/.m2` folder survived from one build to the next.

```
podman run --rm cache-demo ls /root/.m2
```

Observe `ls: /root/.m2: No such file or directory`: the cache is **not** in the image.

Explanation. The *cache mount* is a folder Buildah keeps in your user storage and mounts into the build container for the duration of a single instruction. On a real project, `RUN --mount=type=cache,target=/root/.m2 mvn package` saves you from re-downloading 300 MB of dependencies on every build — even when `pom.xml` changes. No `# syntax=` line was needed: Buildah understands `--mount` natively.

Step 6 — musl or glibc?

```
podman run --rm --entrypoint sh docker.io/library/eclipse-temurin:21-jre-alpine -c 'ldd --
version 2>&1 | head -1; head -1 /etc/os-release'
podman run --rm --entrypoint sh docker.io/library/eclipse-temurin:21-jre -c 'ldd --version 2>&1
| head -1; head -1 /etc/os-release'
```

Observe `musl libc (x86_64)` / `Alpine Linux` on one side, `ldd (Ubuntu GLIBC 2.xx)` / `Ubuntu` on the other.

```
podman run --rm --entrypoint sh docker.io/library/eclipse-temurin:21-jre-alpine -c 'apk info |
wc -l'
podman run --rm --entrypoint sh docker.io/library/eclipse-temurin:21-jre -c 'dpkg -l | grep -c
^ii'
```

Observe about `73` packages versus `140`.

Explanation. Run this command **before** you migrate an image to Alpine: a native library compiled for `glibc` will not load with `musl`. The package count, meanwhile, is what a vulnerability scanner counts: half the packages means half the potential CVEs.

Step 7 — Distroless: no shell at all

```
podman pull gcr.io/distroless/java21-debian12
podman images --format '{{.Repository}} {{.Size}}' | grep distroless
```

Observe `194 MB` — smaller than the Alpine JRE, even though this is Debian.

Create `Dockerfile.distroless`:

```
FROM gcr.io/distroless/java21-debian12
COPY --from=localhost/api-multi:1.0 /app/api.jar /app/api.jar
ENTRYPOINT ["java", "-jar", "/app/api.jar"]
```

```
podman build -f Dockerfile.distroless -t api-distroless:1.0 .
podman run -d --name d -p 18082:8080 api-distroless:1.0
sleep 2 ; curl -s localhost:18082/actuator/health ; echo
podman exec d sh -c 'ls'
podman exec d id
podman rm -f -t 0 d
```

Observe `{"status": "UP"}` — the API is running — then `executable file sh not found in $PATH`, and the same error for `id`: this image contains **nothing** besides Java and your JAR.

Explanation. `COPY --from=` also accepts the name of an existing **image**, not just a stage. And an image with no shell gives an attacker who gains code execution neither `sh`, nor `curl`, nor `apt` — but it locks you out too. You compensate with rich logs, `/actuator`, and `podman cp` to extract a file.

Clean-up

```
podman rmi api-mono:1.0 api-multi:1.0 web-multi:1.0 cache-demo api-distroless:1.0 \
    gcr.io/distroless/java21-debian12 docker.io/library/node:22-alpine 2>/dev/null
podman rmi $(podman images --filter dangling=true -q) 2>/dev/null
podman images --format '{{.Repository}}:{{.Tag}}'
```

Observe that `alpine`, `nginx:alpine`, `eclipse-temurin:21-jdk`, `eclipse-temurin:21-jre` and `eclipse-temurin:21-jre-alpine` remain. Keep `~/labo-docker/05/Dockerfile`: the stack in the upcoming labs will use it.

What you must be able to state now

- A single-stage image weighs as much as its tooling: 488 MB for a 2 KB JAR.

- Multi-stage keeps only the copied artefact: 209 MB, with no sources and no compiler — and `--target` lets you inspect a stage.
- The Angular front end in production is a 64 MB nginx image; Node is not in it.
- A *cache mount* survives builds and never enters the image; Buildah handles it without `#syntax=`.
- Alpine = `musl`, Ubuntu = `glibc`: `ldd --version` tells you, and a test confirms it.
- Distroless: `{"status":"UP"}` but no `sh` — security traded against observability.